

Thes1s Agent Architecture

Complete AI Analyst Team Breakdown

Date: March 29, 2026

Version: Pre-v1.2 (Full Story Milestone)

AI-Powered Rule One Investment Research

9 Specialist Agents | 3-Stage Gated Workflow

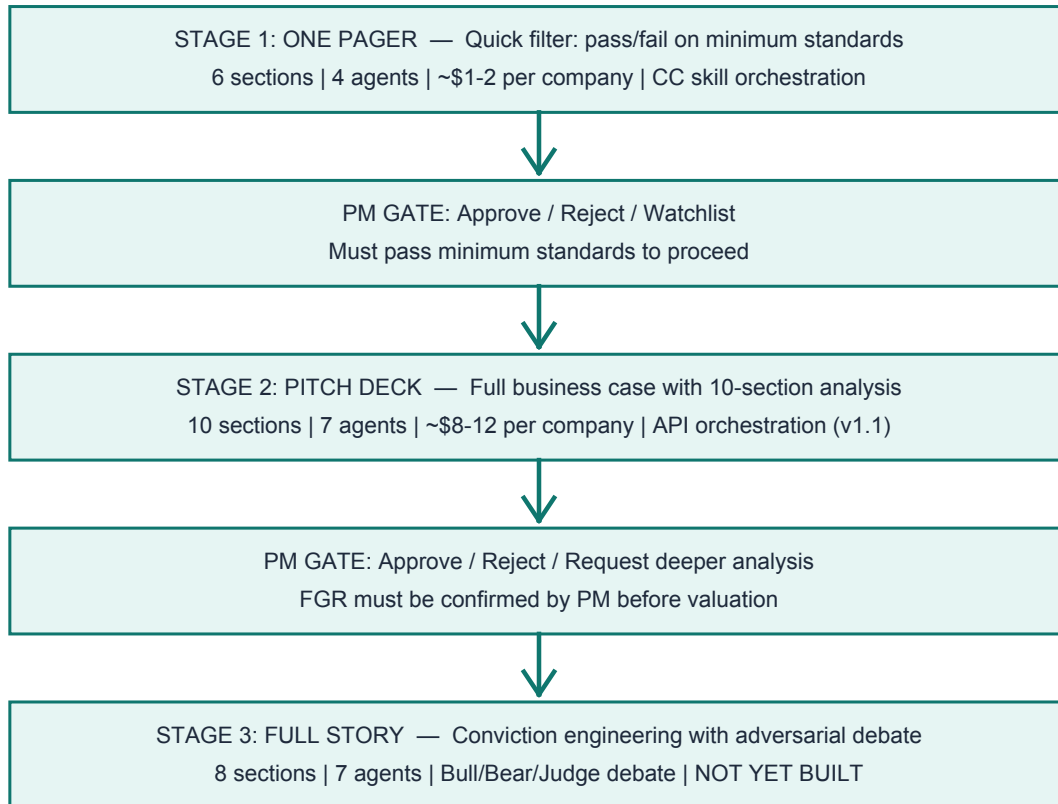
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1. 3-Stage Gated Workflow Overview

Thes1s follows a 3-stage gated research workflow inspired by Rule One investing. Each stage is a filter — the user (Portfolio Manager) must approve before the next stage unlocks. The stages increase in depth and conviction:

Research Pipeline — 3 Gated Stages



Key Principle: Inheritance

Each stage inherits ALL findings from the previous stage. The Full Story doesn't redo the Pitch Deck — it deepens it. Pitch Deck findings flow into Full Story agents as prior context. The DataPacket is also refreshed with any new data.

2. Agent Roster — The Analyst Team

9 AI specialist agents + 1 code-driven assembler. Each agent has a focused role, specific curriculum files, and a defined DataPacket slice. No agent sees all the data — they get exactly what they need for their analysis.

Analysis Agents (7)

Agent	Role	Model	Stages Used
business-analyst	Qualitative business evaluator — moat, simplicity, meaning	Sonnet	OP, PD, FS
financial-analyst	Quantitative expert — growth, returns, FCF, debt, balance..	Sonnet	OP, PD, FS
valuation-specialist	Valuation — FGR derivation, MOS, PBT, Ten Cap, Equity ..	Opus	OP, PD, FS
competitor-evaluator	Moat validation — peer landscape, competitive positioning	Sonnet	PD, FS
management-evaluator	Management quality — compensation, insiders, guru own..	Sonnet	PD, FS
risk-analyst	Risk identification — PEST, events, red flags, bear cases	Opus	PD, FS
synthesis-writer	Final verdict — Buffett-style prose, overall thesis assembly	Opus	OP, PD, FS

OP = One Pager | PD = Pitch Deck | FS = Full Story

Preprocessing Agents (3)

Agent	Role	Model	Used In
annual-reader	Extracts key insights from 10-K filings (full-year data)	Sonnet	PD, FS
quarterly-reader	Extracts recent quarterly updates from 10-Q filings	Sonnet	PD, FS
primary-source-reader	Deep filing/transcript analysis (Opus-level reasoning)	Opus	PD, FS

PSR agents run before analysis agents. They read SEC filings and earnings call transcripts, then pass their findings to downstream agents as context. This lets analysis agents focus on judgment, not extraction.

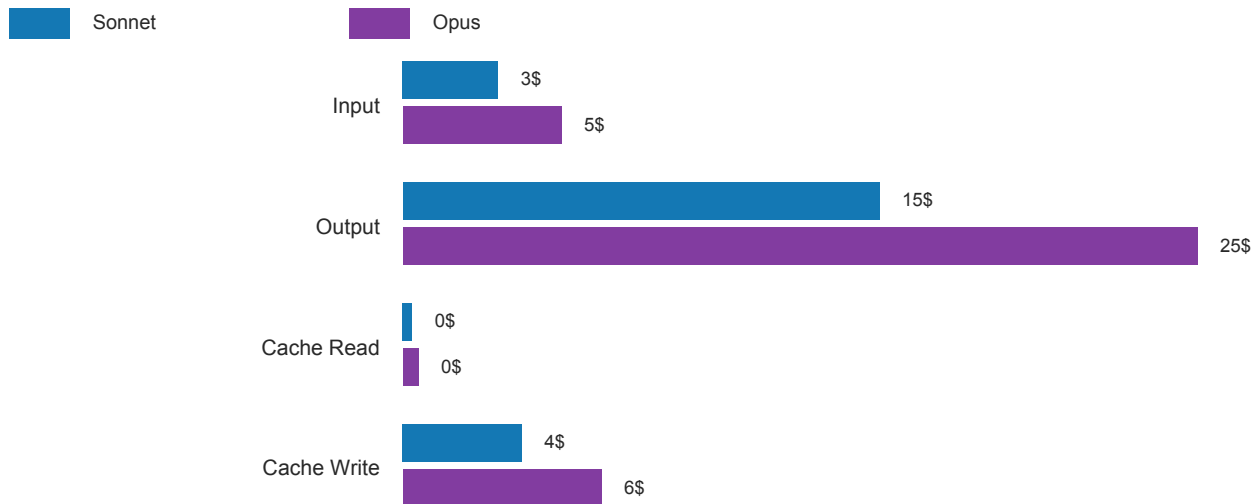
Non-AI Components (1)

Component	Role	Implementation
data-assembler	DataPacket assembly from 20+ data engines	Pure code (nodeAdapter.js + dataExport.js)

3. Agent Model Assignment & Cost Profile

Agents use either Sonnet (fast, cheap) or Opus (deep reasoning, expensive). The assignment follows a simple rule: Opus where judgment matters, Sonnet where thoroughness matters.

Token Cost per 1M Tokens — Sonnet vs Opus



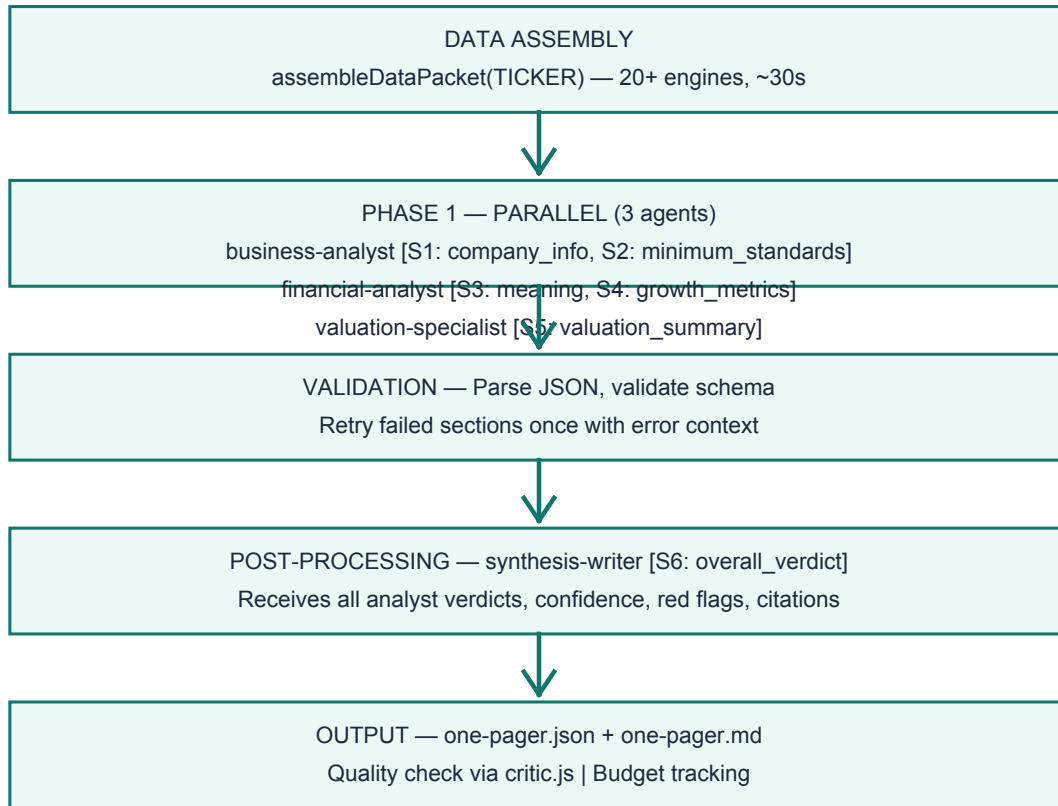
Why Opus for These Agents?

Agent	Reasoning
valuation-specialist	FGR derivation requires weighing 5 inputs + judgment calls. Getting valuation wrong = bad..
risk-analyst	PEST analysis and event assessment need nuanced reasoning about macro trends and st..
synthesis-writer	Final verdict must synthesize all agent findings into coherent Buffett-style prose with convic..
primary-source-reader	Deep reading of 10-K filings requires Opus-level comprehension to extract non-obvious ins..

4. One Pager Pipeline (Stage 1)

The One Pager is a quick filter. It answers: "Does this company pass minimum standards?" Uses 4 agents in 2 phases. Currently runs as a Claude Code skill (not API). Single-agent design — no inter-agent communication needed at this stage.

One Pager Execution Flow



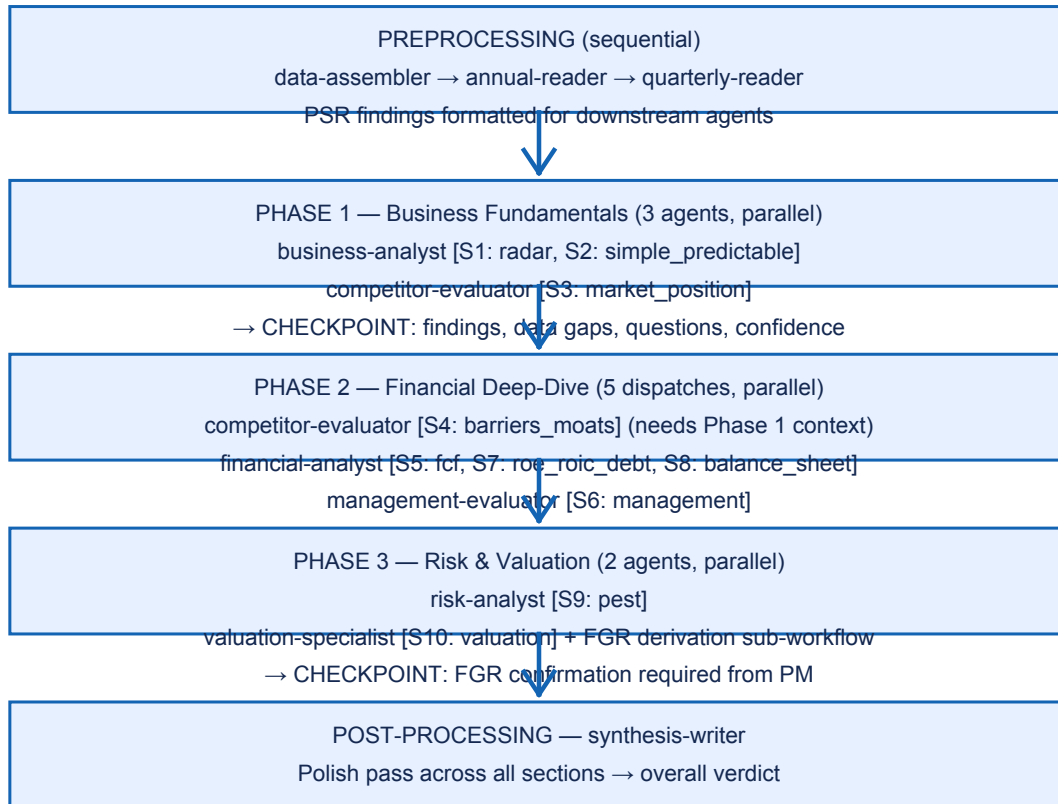
One Pager Section Map

#	Section Key	Agent	Model	What It Produces
1	company_info	business-analyst	Sonnet	Company snapshot, industry, revenue sources
2	minimum_standards	business-analyst	Sonnet	Pass/fail on Rule One minimums
3	meaning	financial-analyst	Sonnet	Business understanding, KPIs
4	growth_metrics	financial-analyst	Sonnet	Big 4 growth rates, consistency
5	valuation_summary	valuation-specialist	Opus	Quick MOS/PBT check
6	overall_verdict	synthesis-writer	Opus	PASS / FAIL / WATCHLIST verdict

5. Pitch Deck Pipeline (Stage 2)

The Pitch Deck is a full 10-part business case. It uses 7 agents across 3 phases plus preprocessing. Currently runs via Claude API (migrated from CC in v1.1). Proven at \$8.53/company, 19-minute runtime, 94/100 mechanical quality.

Pitch Deck Execution Flow



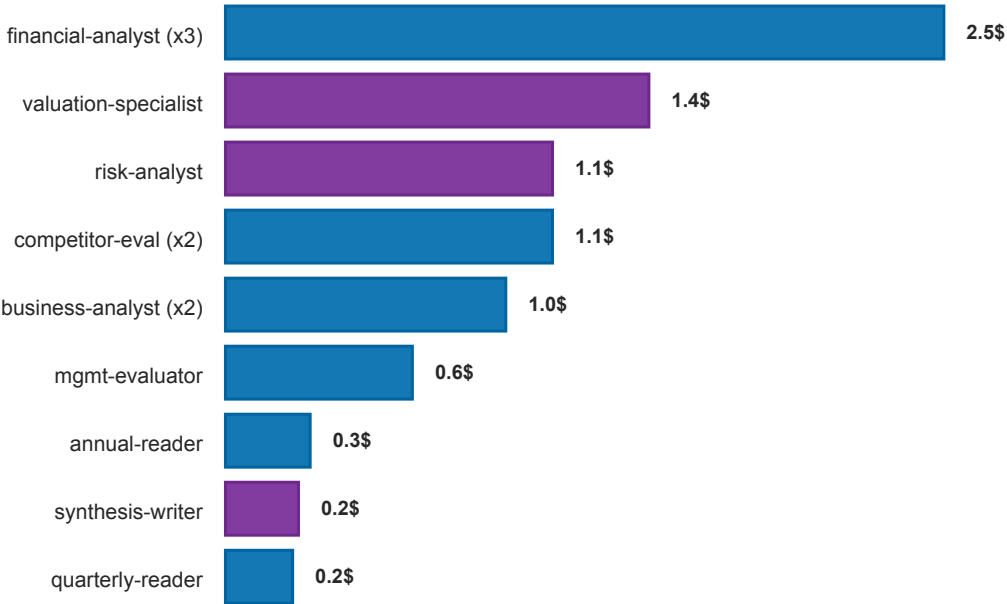
Pitch Deck Section Map

#	Section Key	Agent	Model	Phase
1	radar	business-analyst	Sonnet	1
2	simple_predictable	business-analyst	Sonnet	1
3	market_position	competitor-evaluator	Sonnet	1
4	barriers_moats	competitor-evaluator	Sonnet	2
5	fcf	financial-analyst	Sonnet	2
6	management	management-evaluator	Sonnet	2
7	roe_roic_debt	financial-analyst	Sonnet	2

#	Section Key	Agent	Model	Phase
8	balance_sheet	financial-analyst	Sonnet	2
9	pest	risk-analyst	Opus	3
10	valuation	valuation-specialist	Opus	3

SFM V4 Live Run Results (Phase 10)

Cost per Agent (\$USD) — SFM Pipeline Run

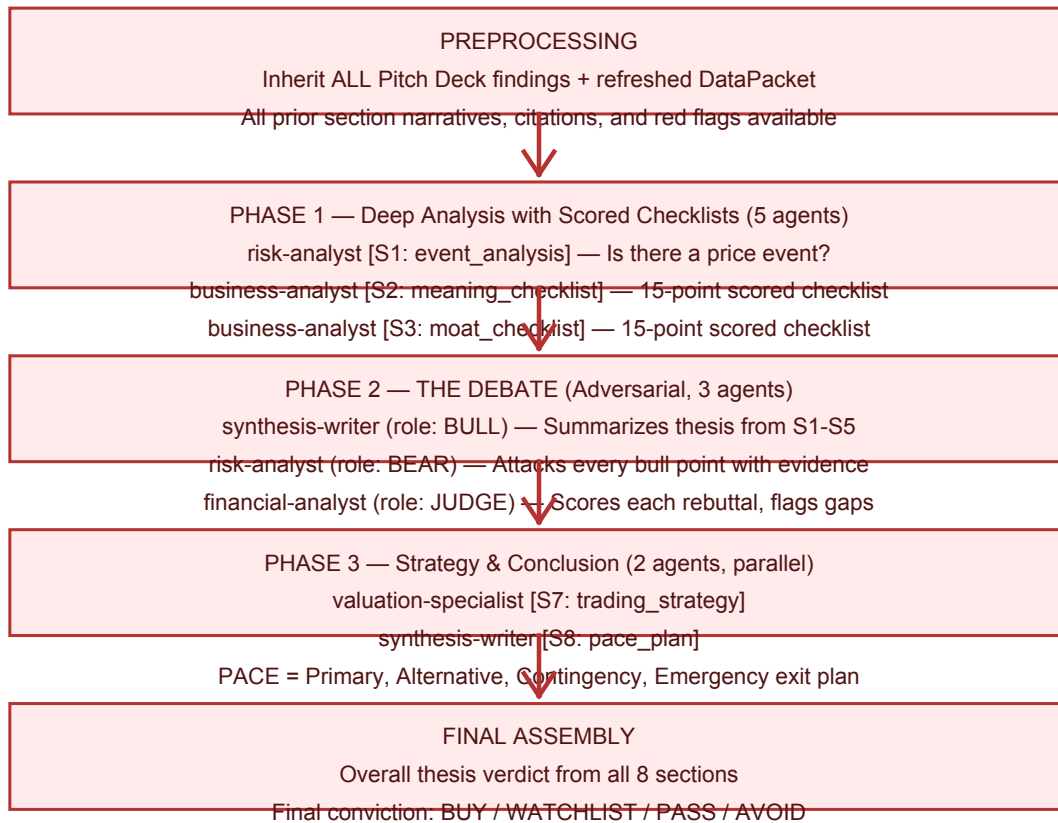


Total: \$8.53 | Runtime: 19 min | Quality: 94 mechanical / 93 methodology

6. Full Story Pipeline (Stage 3) — NOT YET BUILT

The Full Story is "conviction engineering." It is the final gate before capital deployment. It includes 43-item scored checklists, an adversarial Bull/Bear/Judge debate, and a complete trading + PACE strategy. The dispatch table is defined but NO implementation exists yet.

Full Story Planned Execution Flow



Full Story Section Map

#	Section Key	Agent	Model	Phase	Checklist Points
1	event_analysis	risk-analyst	Opus	1	—
2	meaning_checklist	business-analyst	Sonnet	1	15 items
3	moat_checklist	business-analyst / competitor..	Sonnet	1	15 items
4	management_checklist	management-evaluator	Sonnet	1	13 items
5	valuation_confirmation	valuation-specialist	Opus	1	—
6	inversion_rebuttal	bull + bear + judge (debate)	Opus	2	—

#	Section Key	Agent	Model	Phase	Checklist Points
7	trading_strategy	valuation-specialist	Opus	3	—
8	pace_plan	synthesis-writer	Opus	3	—

The Debate — Phase 2 Detail

Phase 2 is unique: three agents take adversarial roles on the SAME section (S6). This is inversion and rebuttal — a core Rule One concept. The Bull summarizes the thesis, the Bear attacks it with evidence, and the Judge scores each rebuttal for strength and identifies gaps in reasoning.

Role	Agent	Responsibility
BULL	synthesis-writer	Compile strongest case from all prior sections. Cite specific data.
BEAR	risk-analyst	Attack each bull point with counter-evidence. Web search for bear theses.
JUDGE	financial-analyst	Score each rebuttal (strong/weak/inconclusive). Identify unaddressed risks.

6b. Token & Context Budget per Agent

Every time an agent is spawned, its context window is loaded with: (1) system message (universal context + PSR findings + agent prompt + curriculum), (2) user message (field path block + DataPacket slice + section assignment + prior sections + PM feedback). Here is the breakdown of what each agent costs in context before it even starts thinking.

System Message Components (fixed cost per agent)

Component	Size (chars)	Est. Tokens	Cached?	Notes
Universal Context	18,349	~4,600	Yes	rule-one-fundamentals + tools-for-analysis
PSR Findings	varies	~2,000-8,000	Yes	Annual + quarterly reader output
Agent Prompt	22-40K	~5,500-10,000	No	Varies by agent (see below)
Curriculum Files	0-63K	~0-15,700	No	Methodology files per agent role

Cached components (universal context, PSR findings) are marked `cache_control: ephemeral`. They count toward input tokens only once per pipeline run — subsequent agents pay the much cheaper `cache_read` rate (\$0.30/1M for Sonnet, \$0.50/1M for Opus).

Per-Agent Context Breakdown (System Message Only)

Agent	Prompt	Curriculum	Fixed Conte..	Total System
business-analyst	~8,000	~10,600	~4,600	~23,200 tokens
financial-analyst	~9,500	~5,900	~4,600	~20,000 tokens
valuation-specialist	~7,800	~15,700	~4,600	~28,100 tokens
competitor-evaluator	~8,800	~10,700	~4,600	~24,100 tokens
management-evaluator	~9,600	~6,200*	~4,600	~20,400 tokens
risk-analyst	~10,100	~9,100	~4,600	~23,800 tokens
synthesis-writer	~5,600	~0*	~4,600	~10,200 tokens
annual-reader	~7,300	0	~4,600	~11,900 tokens
quarterly-reader	~7,600	0	~4,600	~12,200 tokens

* management-evaluator and synthesis-writer reference `buffett_letters_claude_training_set/` directory in their config, but `loadCurriculum()` reads files — not directories. This reference silently fails in the API pipeline. The Buffett style comes from the prompt text instead. This is a gap to address in v1.2 — either load specific letter excerpts or remove the reference.

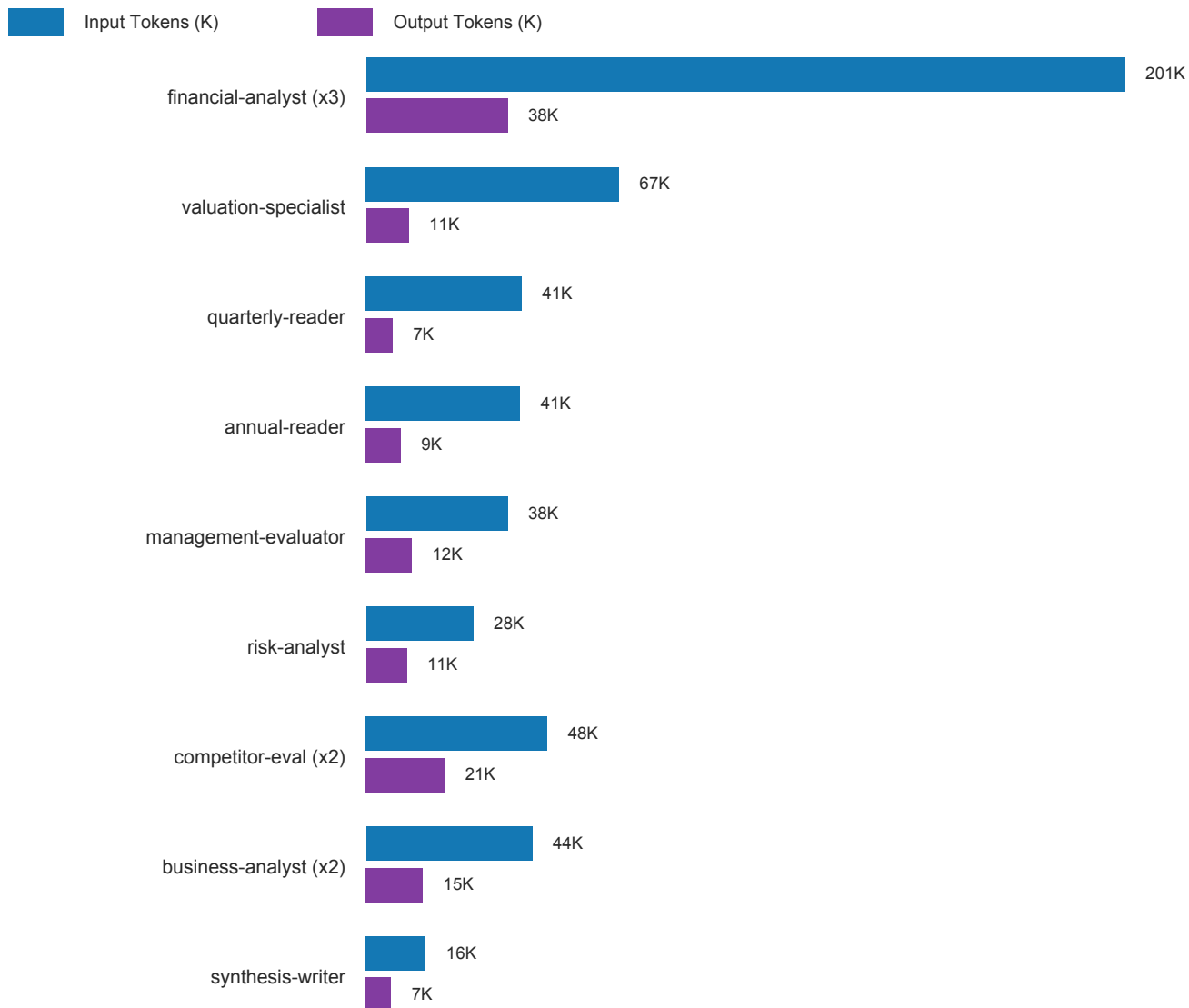
User Message Components (varies by company)

Component	Typical Size	Notes
Field Path Block	~500-1,500 tokens	2-level depth, 20-key cap per top-level field
DataPacket Slice	~2,000-40,000 tokens	financial-analyst gets largest slice (7 fields)
Section Assignment	~200-500 tokens	Which sections to generate, format instructions
Prior Section Context	~0-15,000 tokens	Phase 2+ gets Phase 1 findings as context
PM Feedback	~0-500 tokens	From checkpoint reviews (if any)

SFM Live Run — Actual Token Usage per Agent

From the Phase 10 live SFM pipeline run (\$8.53 total). These are actual API usage numbers, not estimates. Total: 518,907 input + 129,668 output = 648,575 tokens.

Actual Token Usage — SFM Pipeline Run (thousands)



Prompt Cache Performance

Cache hit rate on SFM: 53% (938K cache_read tokens out of 1.77M total input). Target is 70%. The three-block system message structure (universal context + PSR findings + agent-specific) enables caching of shared content across agents. The 53% rate was below target because filing content was not yet included in the cached DataPacket.

7. DataPacket Assembly & Slicing

The DataPacket is the canonical data source for all agents. Assembled from 20+ engines (EDGAR XBRL, Yahoo Finance, Finviz, SEC filings, etc.), it contains everything an agent might need. Each agent receives only its slice — never the full packet.

DataPacket Assembly Pipeline



DataPacket Slicing per Agent

Each agent receives only the DataPacket fields listed in its config. This focuses the agent's context and prevents cross-contamination.

Agent	DataPacket Slice
business-analyst	companyInfo, classification, ruleOneScore, peers
financial-analyst	financials, ttm, growthRates, returnMetrics, debtMetrics, fcf, keyMetrics
valuation-specialist	growthRates, returnMetrics, fcf, analystEstimates, ttm, currentPrice, keyMetrics
competitor-evaluator	peers, peerMetrics, classification, companyInfo
management-evaluator	compensation, insiders, gurus, companyInfo
risk-analyst	companyInfo, events, analystEstimates, classification
synthesis-writer	(none — receives analyst summaries instead)
annual-reader	companyInfo, classification, financials, ttm, filings, filingContent
quarterly-reader	companyInfo, classification, financials, ttm, filings, filingContent, transcripts

8. Curriculum & Context Loading

Every agent receives: (1) Universal context — Rule One fundamentals + tools, (2) Agent-specific curriculum — methodology files relevant to their role, (3) PSR findings — insights from preprocessing agents. Universal context is cached across agents for cost savings.

Universal Context (All Agents)

File	Content
rule-one-fundamentals.md	Investment philosophy, compound returns, margin of safety, event investing, ..
tools-for-analysis.md	Financial formulas, data source commands, calculation examples

These are cached (cache_control: ephemeral) so they only count toward input tokens once per pipeline run.

Agent-Specific Curriculum

Agent	Curriculum Files	Why
business-analyst	pitch-deck-I, one-pager, story-form-I, advanced-..	Moat definitions, business evaluation, mea..
financial-analyst	advanced-financial-analysis, fgr, capex-cash-flo..	Financial formulas, growth rate methodolo..
valuation-specialist	pitch-deck-IV, fgr, equity-bond-research, advanc..	All 4 valuation methods, FGR derivation, B..
competitor-evaluator	pitch-deck-I, pitch-deck-II, story-form-I, advance..	Competitive positioning, moat validation fra..
management-evaluator	pitch-deck-II, advanced-financial, buffett_letters/..	Leadership evaluation, compensation anal..
risk-analyst	pitch-deck-III, story-form-II, advanced-financial, ..	PEST analysis, inversion methodology, ev..
synthesis-writer	buffett_letters_claude_training_set/	Buffett writing style for prose quality
PSR agents (3)	(none — universal context only)	Raw extraction, not analysis

Contamination Boundary

CRITICAL: Agents must NEVER access example files. These directories are excluded from all agent configs:

- knowledge/stage-1-one-pager/examples/ (LULU One Pager)
- knowledge/stage-2-pitch-deck/examples/ (LULU Pitch Deck)
- knowledge/stage-3-full-story/examples/ (LULU Full Story)
- knowledge/pre-course-examples/ (EW, SFM, MU, ODFL)

These exist only for the PM to compare output quality after generation. If agents see examples, they pattern-match instead of independently analyzing.

9. Section-to-Agent Mapping (All Stages)

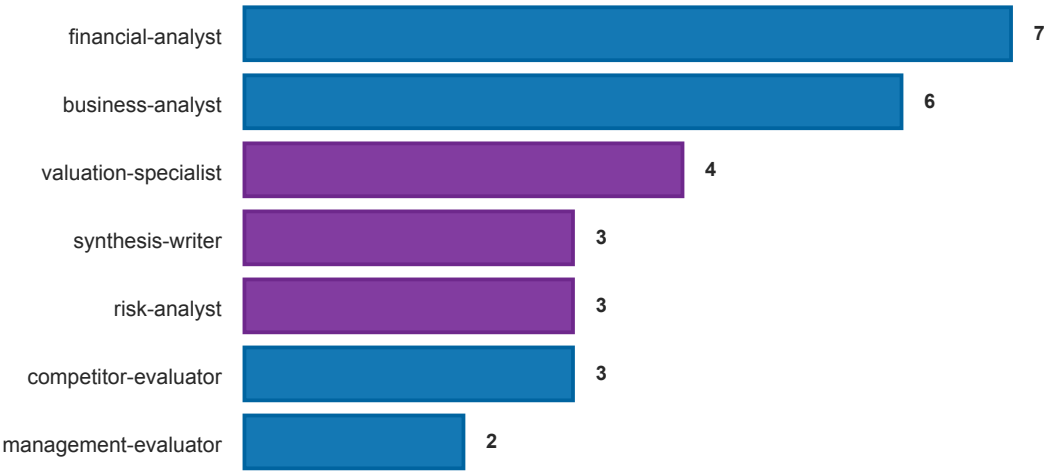
Complete mapping from the orchestrator config. Shows which agent handles which section number in each stage. Agents are reused across stages with different section assignments.

Cross-Stage Agent Usage

Agent	One Pager Sections	Pitch Deck Sections	Full Story Sections
business-analyst	S1, S2	S1, S2	S2 (meaning), S3 (moat)
financial-analyst	S3, S4	S5, S7, S8	S5 (valuation), S6 (judge)
valuation-specialist	S5	S10	S5, S7 (trading strategy)
competitor-evaluator	—	S3, S4	S3 (moat checklist)
management-evaluator	—	S6	S4 (management checklist)
risk-analyst	—	S9	S1 (events), S6 (bear)
synthesis-writer	S6	(polish pass)	S6 (bull), S8 (PACE plan)

Agent Reuse Across Stages

Number of Section Assignments per Agent (All 3 Stages)



10. Current Status & Gaps for Full Story

What EXISTS Today

- All 10 agent configs (config.json) — roles, models, curriculum, DataPacket slices defined
- All 10 agent prompts (prompt.md) — investigation mandate, web research, curriculum sections
- Dispatch table — Full Story workflow defined in dispatch-table.json (3 phases + debate)
- Section keys defined: event_analysis, meaning_checklist, moat_checklist, management_checklist, valuation_confirmation, inversion_rebuttal, trading_strategy, pace_plan
- ReportSectionSchema — output format already defined and validated
- API dispatch engine (aiResearch.js) — proven on Pitch Deck, ready for Full Story
- Pipeline manager (pipelineManager.js) — wave orchestration, checkpoints, budget tracking
- Quality system (critic.js) — mechanical + methodology scoring
- Curriculum files — story-form-I.md (sections 1-4) and story-form-II.md (sections 5-8)

What NEEDS to Be Built

- Full Story CC skill orchestration — like generate-one-pager but for Stage 3
- Checklist scoring mechanism — 43-item checklists (15+15+13) need scored output format
- Debate orchestration — Bull/Bear/Judge sequential dispatch with role switching
- Pitch Deck inheritance — loading prior PD findings as context for FS agents
- Full Story methodology checks in critic.js — new curriculum-derived checks
- Trading strategy + PACE plan prompt engineering — new section types
- Growth ceiling analysis — market share ceiling validation
- Inversion protocol — structured "for every reason, create counter-argument"

Key Questions for v1.2 Planning

1. Are the existing agent prompts sufficient for Full Story, or do they need Full Story-specific prompt sections?
2. The dispatch table shows competitor-evaluator on Full Story S3 (moat_checklist) but the orchestrator mapping shows business-analyst. Which is correct?
3. Phase 2 debate: should Bull → Bear → Judge be strictly sequential, or can Bull and Bear run in parallel with Judge receiving both?
4. Do we need new agents for the Full Story, or just new prompts for existing ones?
5. How should checklist items be scored? Binary pass/fail? 1-5 scale? Narrative + score?